

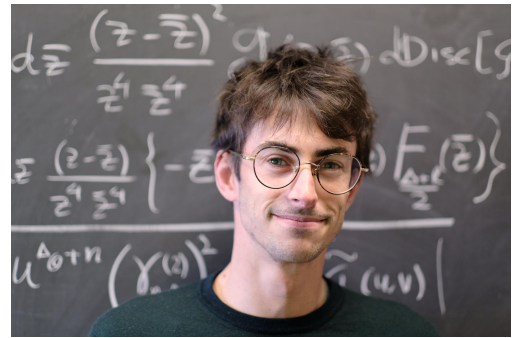
Andrea Trost

Observational Cosmologist – OATs

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Profile

- ◇ I am an observational cosmologist, currently working in INAF-OATs in Trieste. My research mainly revolves around the analysis of absorption lines in quasar spectra as a probe of cosmology, astroparticle physics and the physical state of the intergalactic medium. I have experience in handling both cosmological simulation data products and observational datasets.
- ◇ I was born in Tolmezzo (Udine) on November 6th, 1998, and grew up in Udine, in north-eastern Italy.
- ◇ Currently, I am exploiting high-resolution high-signal-to-noise VLT/ESPRESSO¹ spectra to detect the expansion of the Universe in real time and constrain the mass of warm dark matter by measuring the Ly- α flux power spectrum at the smallest scales within the EQUALS collaboration².

Employment & Education

- 2026 – present ◇ **Postdoctoral fellow** – INAF - Osservatorio Astronomico di Trieste, Trieste, Italy.
(incarico di ricerca)
- 2025 – 2026 ◇ **Postdoctoral fellow** – SISSA, Trieste, Italy.
Borsa per attività di ricerca e perfezionamento – Advisor: M. Viel
- 2022 – 2025 ◇ **PhD in Physics – cum laude** – University of Trieste, Italy.
Thesis title: *The Lyman- α Forest as a Probe of Fundamental Cosmology: Detecting Dark Photon Dark Matter and Measuring the expansion of the Universe*
Supervisor: S. Cristiani; Defence 9th Feb 2026
- 2020 – 2022 ◇ **Master's Degree in Physics, curriculum of Astrophysics and Cosmology** – University of Trieste, Italy.
Thesis title: *Probing the small scale structure of the IGM with lensed quasar spectra.*
Supervisor: S. Cristiani – Grade: 110/110 *cum laude*
- 2017 – 2020 ◇ **Bachelor's Degree in Physics** – University of Trieste, Italy.
Thesis title: *CUBES: simulation and analysis of ionizing escape fraction from high redshift star-forming objects.*
Supervisor: S. Cristiani – Grade: 109/110

Skills

- Languages ◇ Italian (native), English (fluent), Portuguese (basic)
- Coding ◇ Python, IDL, Fortran, C, Arduino, HTML, $\text{\LaTeX}$.
- Astronomical Softwares ◇ Astrocook³, QFitsView⁴, Marz⁵, EsoReflex⁶, DS9, GAIA, IRAF, VPFIT.

¹ <https://www.eso.org/sci/facilities/paranal/instruments/espresso.html>

² Berg, T., et al. *The Messenger* 195 (September 2025): <https://doi.org/10.18727/0722-6691/5395>

³ <https://das-oats.github.io/astrocook/>

⁴ <https://www.mpe.mpg.de/~ott/dpuser/qfitsview.html>

⁵ <https://samreay.github.io/Marz/#/overview>

⁶ <https://www.eso.org/sci/software/esoreflex/>

Publications

Journal Articles

- 1 Porru, M. et al. (Apr. 2026). “Estimating the completeness of the QUBRICS survey with 3501 quasi-stellar object redshifts from Gaia DR3 spectra”. *Astron. Astrophys.* 708, A363.  DOI: 10.1051/0004-6361/202558149. arXiv: 2603.06907 [astro-ph.CO].
- 2 Noterdaeme, P. et al. (Mar. 2026). “One H₂ molecule per ten million H atoms reveals sub-parsec-scale cold overdensities at $z \sim 4$ ”. *Astron. Astrophys.* 707, A66, A66.  DOI: 10.1051/0004-6361/202558667.
- 3 **Trost, A.**, C. M. J. Marques, et al. (Mar. 2026). “The ESPRESSO Redshift Drift Experiment III – The Third Epoch of QSO J052915.80-435152.0”. arXiv: 2603.02318 [astro-ph.CO].
- 4 Marques, C. M. J., **A. Trost**, et al. (Dec. 2025). “The ESPRESSO Redshift Drift Experiment. II. High-Fidelity Spectral Model of QSO J212540.96-171951.32’s Lyman- α forest”. *In review*.
- 5 Di Stefano, S., V. D’Odorico, et al. (Oct. 2025). “Detecting the IGM metal enrichment with the 2-point correlation function of the flux. Application to the UVES deep spectrum”. *Astron. Astrophys.*  DOI: 10.1051/0004-6361/202556252.
- 6 **Trost, A.**, C. M. J. Marques, et al. (July 2025). “The ESPRESSO Redshift Drift Experiment - I. High-resolution spectra of the Lyman- α forest of QSO J052915.80-435152.0”. *Astron. Astrophys.* 699, A159.  DOI: 10.1051/0004-6361/202554502.
- 7 **Trost, A.**, J. S. Bolton, et al. (Apr. 2025). “Constraints on dark photon dark matter from Lyman- α forest simulations and an ultrahigh signal-to-noise quasar spectrum”. *Phys. Rev. D* 111 (8), p. 083034.  DOI: 10.1103/PhysRevD.111.083034.
- 8 Grazian, A. et al. (Oct. 2024). “What Are the Pillars of Reionization? Revising the AGN Luminosity Function at $z \sim 5$ ”. *The Astrophysical Journal* 974.1, 84, p. 84.  DOI: 10.3847/1538-4357/ad6980.
- 9 Cristiani, S., G. Cupani, et al. (Mar. 2024). “Probing the small-scale structure of the intergalactic medium with ESPRESSO: spectroscopy of the lensed QSO UM673”. *Monthly Notices of the Royal Astronomical Society* 528.4, pp. 6845–6860.  DOI: 10.1093/mnras/stae432.
- 10 Guarneri, F. et al. (Feb. 2024). “Fundamental physics with ESPRESSO: a new determination of the D/H ratio towards PKS1937-101”. *Monthly Notices of the Royal Astronomical Society*.  DOI: 10.1093/mnras/stae452.
- 11 Cristiani, S., M. Porru, et al. (Apr. 2023). “Spectroscopy of QUBRICS quasar candidates: 1672 new redshifts and a Golden Sample for the Sandage Test of the Redshift Drift”. *Monthly Notices of the Royal Astronomical Society*. ISSN: 0035-8711.  DOI: 10.1093/mnras/stad1007.
- 12 Evans, C. et al. (Oct. 2022). “The CUBES science case”. *Experimental Astronomy*.  DOI: 10.1007/s10686-022-09864-7.
- 13 Cristiani, S., J. M. Alcalá, et al. (Sept. 2022). “CUBES, the Cassegrain U-Band Efficient Spectrograph for the VLT”. *The Messenger* 188, pp. 36–41.  DOI: 10.18727/0722-6691/5278.
- 14 Zanutta, A. et al. (May 2022). “CUBES phase A design overview”. *Experimental Astronomy*.  DOI: 10.1007/s10686-022-09837-w.
- 15 Genoni, M. et al. (Mar. 2022). “The CUBES instrument model and simulation tools”. *Experimental Astronomy*.  DOI: 10.1007/s10686-022-09844-x.

Conference Proceedings

- 1 **Trost, A.** (Feb. 2026). “Measuring the Cosmic Redshift Drift with the VLT: A Long-Term ESPRESSO Program into the ELT Era”. *VLT Beyond 2030*, 21, p. 21.  DOI: 10.5281/zenodo.18804156.

Experience

Research

- 2025 – present  Within the EQUALS teams, and in close collaboration with Vid Iršić, I am developing a pipeline to measure the Ly- α flux power spectrum at $z \sim 3-4$ using 25 ultra-high-resolution VLT/ESPRESSO quasar spectra, targeting the smallest scales to unveil the suppression signature expected from warm dark matter and other light relics.
- 2023 – present  I lead the ESPRESSO Redshift Drift Experiment, an observational campaign using the Ly- α forest of the brightest known quasar to perform the Sandage-Loeb test, a direct, model-independent probe of cosmic expansion via the tiny temporal drift in absorption-line redshifts. I have characterized the dominant astrophysical and instrumental systematics and delivered the tightest redshift-drift constraints to date, establishing the temporal baseline and integration time required for a significant detection with the forthcoming ELT/ANDES spectrograph.
- 2023 – 2024  I compared real and mock quasar spectra from high-resolution hydrodynamical IGM simulations populated with Dark Photon Dark Matter, exploiting the forest's sensitivity to anomalous heating to place the strongest constraints to date on this candidate's mass and kinetic mixing at $z \sim 2.7$. Here's a brief presentation: 
- 2021 – 2023  I analysed VLT/ESPRESSO spectra of the lensed quasar UM673, comparing absorption features across its two images to constrain the distribution, dimensions, and dynamical properties of metal and neutral hydrogen absorbers in the IGM.
- 2020  I contributed to the VLT/CUBES spectrograph Exposure Time Calculator during the project's phase A, used to assess the instrument's capabilities for cosmic reionization studies.

Astronomical Observations

- Sept. 2026  **4 nights NTT/EFOSC2** in Visitor Mode under run 117.2AF5 "Crossing the QSO Redshift Desert at $z \sim 5.5$ with Machine Learning selection" (PI Grazian).
- Jan.-Mar 2026  **PI of 19.5 h VLT/ESPRESSO** in Service Mode under run 116.28U4 "Breaking the 1 m/s/yr barrier in the measurement of the cosmic redshift drift".
- Apr.-May 2025  **PI of 19.2 h VLT/ESPRESSO** in Service Mode under run 115.28E05 "ESPRESSO in the sky with diamonds: the small scale structure of the IGM with spectroscopy of lensed quasars".
- Jun. 2024  **5.5 nights NTT/EFOSC2** in Visitor Mode under run 113.26R5 "Machine Learning tools to find the Brightest high- z QSOs in the Southern Hemisphere" (PI Guarneri).
- 2023 – present  **Co-I** of 9 accepted observational proposals for ESO telescopes (340 hours in total)
- 2025 – present  **Referee for Observational Proposals:** Telescopio Nazionale Galileo (TNG), Large Binocular Telescope (LBT), and Rapid Eye Mount (REM) — evaluation of time allocation proposals **Referee for Journal Articles:** JCAP,

Talks & Seminars

- Sept. 5th 2026  **Invited talk :** *The ESPRESSO Redshift Drift Experiment* at ESO Colloquia, ESO Vitacura Headquarters, Santiago, Chile.
- June 16th 2026  **Contributed talk :** *New robust constraints on Dark Photon Dark Matter from the intergalactic medium* and **Poster:** *Measuring the Cosmic Redshift Drift with the VLT: A Long-Term ESPRESSO Program into the ELT Era* at the 1st BiCoQ conference: "From gravity to particles", Università di Milano Bicocca, Milan, Italy.

Experience (continued)

- May 7th 2026 ♦ **Contributed talk :** *The ESPRESSO Redshift Drift Experiment* at "UniVersum VII", Università Federico II, Naples, Italy.
- Jan. 27nd 2026 ♦ **Invited Panellist :** *Extreme wavelength calibration & Laser Frequency Comb technology* at "VLT beyond 2030", ESO, Munich, Germany.
- May 22nd 2025 ♦ **Contributed talk:** *Probing the Small-Scale Structure of the IGM with Lensed Quasars* at "The galaxy-IGM connection in the first billion years" Focus week, IFPU, Trieste, Italy.
- May 7th 2025 ♦ **Contributed talk:** *New robust constraints on Dark Photon Dark Matter from the intergalactic medium* at the 5th EuCAPT Annual Symposium, CERN (remote).
- Apr. 14th 2025 ♦ **Contributed talk:** *The ESPRESSO Redshift Drift Experiment – High-fidelity spectra of the Lyman forest of QSO J052915.80-435152.0* at IberiCos 2025, University of Coimbra, Portugal.
- Jan. 17th 2024 ♦ **Contributed talk:** *Fundamental Physics in the Lyman- α Forest with CUBES* at the CUBES Science Team Meeting 2024, Osservatorio Astronomico di Capodimonte, Napoli, Italy.
- Jul. 19th 2023 ♦ **Contributed talk:** *Probing the Small-Scale Structure of the IGM with Lensed Quasars* at StEm65, Sexten, Italy.
- Apr. 4th 2023 ♦ **Contributed talk:** *Probing the Small-Scale Structure of the IGM with Lensed Quasars* at IberiCos 2023, Ponte de Lima, Portugal.
- Feb. 8th 2023 ♦ **Contributed talk:** *Probing the Small-Scale Structure of the IGM with Lensed Quasars* at the ESPRESSO Science Team Meeting 2023, Lanzarote, Spain (remote).

Internships & Schools

- Sep. 2024 ♦ **IV Azores School on Observational Cosmology:** *Fundamental cosmology from the ELT and space facilities*, organized by the Instituto de Astrofísica e Ciências do Espaço, Angra do Heroísmo, Azores, Portugal.
- Nov. - Dec. 2022 ♦ **XXXIII Canary Islands Winter School of Astrophysics:** *Overlaps at the boundaries of Astrophysics, Cosmology and Particle Physics*, organized by the Instituto de Astrofísica de Canarias (IAC), Tenerife, Spain.
- 2021 ♦ **Astronomical Observatory of Trieste:** Beta tester and junior developer for Astrocook's spectrum analysis pipelines and routines.
- 2019 ♦ **Astronomical Observatory of Trieste:** Characterization and calibration of the OATs' photometric camera at the SVAS telescope. Technical use and drafting of the instrument's user manual.

Awards and Achievements

- Jan. 2026 ♦ **Best Poster Prize:** *Measuring the Cosmic Redshift Drift with the VLT: A Long-Term ESPRESSO Program into the ELT Era* at "VLT beyond 2030", ESO, Munich, Germany.
- Jul. 2022 ♦ First place at the XXXVIII cycle PhD in Physics admission rankings, Department of Physics, University of Trieste.
- May 2022 ♦ **PLANCKS22**, Participation to the international team competition in theoretical physics, Munich, Germany. Placed 19th/39. <http://www.dpg-physik.de/plancks>
- Mar. 2022 ♦ **PREPLANCKS22**, Second place at the Italian national team competition in theoretical physics.

Certifications

- Jan. 2022 ♦ **TOEFL iBT (Home Edition)** – Total score: 103/120

Experience (continued)

Other

- Nov. 2022 – present
 - ◇ Attendee and presenter at the *QSO-IGM* and *One-Plot-Chats* weekly journal clubs held at OATs.
 - ◇ Organizer of the *Meet the Speaker* weekly meetings held at OATs.
- 2023 – 2024
 - ◇ Model for the University of Trieste's 2024 5x1000 and 100th anniversary advertisement campaign.
- 2021 – 2022
 - ◇ Student Representative and member of the Boards of Studies, University of Trieste.
 - ◇ Rock climber, certified mountain skier, avalanche safety + search & rescue trained, experienced alpinist and overall outdoor enthusiast.

References

Prof. Matteo Viel

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